Commutative Algebra Cheat Sheet

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Contents

1 Definitions **2**
- 1.1 Basic Stuff . . . 2
- 1.2 Spectrum of a commutative ring . . . 3
- 1.3 Modules, Tensor Product, and Kähler Differential . . . 3
  - 1.3.1 Modules . . . 3
  - 1.3.2 Tensor Product . . . 3
  - 1.3.3 Differentials . . . 4
- 1.4 Integrality, Valuation Rings . . . 5
  - 1.4.1 Integral Elements and Closure . . . 5
  - 1.4.2 Valuation Rings . . . 6
- 1.5 Noetherian Rings . . . 6
- 1.6 Graded Rings . . . 7

2 Results **7**
- 2.1 Nilradical is the intersection of prime ideals . . . 7
- 2.2 Spec(A) is quasi-compact . . . 7
- 2.3 Tensor Product is Right-exact . . . 8
- 2.4 Structure of modules over PID . . . 8
- 2.5 Ring is local iff non-units form an ideal . . . 8
- 2.6 Nakayama's Lemma . . . 9
- 2.7 Finitely presented flat module over local ring is free . . . 9
- 2.8 Going-up Theorem . . . 9
- 2.9 Artin-Tate Lemma . . . 10
- 2.10 Hilbert's Basis Theorem . . . 10
- 2.11 Nullstellensatz Lemma . . . 11
- 2.12 Noether normalization lemma . . . 11
- 2.13 Chevalley's Theorem . . . 12
- 2.14 Equivalent conditions for separable field extensions . . . 12
- 2.15 Finiteness of integral closure . . . 13
- 2.16 Criterion for a graded ring to be Noetherian . . . 13

1 Definitions

1.1 Basic Stuff

Further assume A is a unital commutative ring.

Definition 1.1. *An element $x \in A$ is said to be:*

- ***unit*** *if $x^{-1} \in A$;*
- ***nilpotent*** *if $x^n = 0$ for some n;*
- ***idempotent*** *if $x^2 = x$.*

Definition 1.2. *$I \subseteq A$ is an **ideal** if $aI \subseteq I$ for every $a \in A$ and I is closed under addition. In particular, $\mathfrak{p} \subsetneq A$ is **prime** if $xy \in \mathfrak{p}$ implies $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$, while $\mathfrak{m} \subsetneq A$ is **maximal** if no ideals are strictly contained between $\mathfrak{m}$ and A.*

Definition 1.3. *The **radical** of an ideal is $\sqrt{I} = \{x \in A : x^n \in I \text{ for some } n\}$. In particular, **nilradical** is $\mathfrak{N}_A = \{x \in A : x^n = 0 \text{ for some } n\}$. Finally, I is **radical** if $I = \sqrt{I}$.*

Definition 1.4. *Let $S \subseteq A$ be a multiplicative system (that is, S is endowed with a closed multiplication map). Then, **localization** $A[S^{-1}]$ (or sometimes notation $S^{-1}A$ is used) can be defined in two ways:*

(i) For $S = \{f_\alpha\}_{\alpha \in I}$, $A[S^{-1}] = A[T_\alpha]_{\alpha \in I}/(T_\alpha f_\alpha - 1)_{\alpha \in I}$.

(ii) The set $\{\frac{a}{b} : a \in A, b \in S\}$ where two elements $\frac{a}{b} \sim \frac{a'}{b'}$ iff exists $s \in S$ such that $s(ab' - a'b) = 0$.

Lemma 1.5. *Two basic facts about $A[S^{-1}]$:*

(i) $\ker(A \to A[S^{-1}]) = \{g \in A : gs = 0 \text{ for some } s \in S\}$.

(ii) Let $I \subseteq A$ be an ideal and let $S \subseteq A$ be a multiplicative subset. Let $\pi : A \to A/I$ be a projection. Then, there exists an isomorphism:

$$(A/I)[\pi(S)^{-1}] \cong A[S^{-1}]/IA[S^{-1}].$$

Definition 1.6. *Map $A \to B$ is said to be **of finite type** if B is a finitely generated as an A-algebra, while it is **finite** if B is a finitely generated A-module. Finally, B is **finitely presented** if $B \cong A[x_1, \dots, x_n]/(f_1, \dots, f_m)$ for some elements $f_1, \dots, f_m \in A[x_1, \dots, x_n]$.*

1.2 Spectrum of a commutative ring

Definition 1.7. *Define a set* $Spec(A) = \{\mathfrak{p} \subseteq A : \mathfrak{p} \text{ is prime}\}$. *We say that a subset* $Z \subseteq Spec(A)$ *is* ***closed*** *if* $Z = V(I)$ *for some ideal* I *and*

$$V(I) = \{\mathfrak{p} \in Spec(A) : I \subseteq \mathfrak{p}\}.$$

Open sets are if turn are defined as $D(I) = \{\mathfrak{p} \in Spec(A) : I \nsubseteq \mathfrak{p}\}$.

Lemma 1.8. *Closed subsets define a topology: they are closed under finite unions and arbitrary intersections, and both* $\emptyset$ *and* $Spec(A)$ *are closed.*

Proof. The last fact is obvious: $\mathrm{Spec}(A) = V(0)$ while $\emptyset = V(A)$. Now consider the union $V(I_0) \cup V(I_1)$. It consists of all prime ideals that contain either I_0 or I_1. Thus, $V(I_0) \cup V(I_1) = V(I_0 \cap I_1) = V(I_0 I_1)$. It is also easy to see that $V(\sum I_\alpha) = \bigcap V(I_\alpha)$. □

1.3 Modules, Tensor Product, and Kähler Differential

1.3.1 Modules

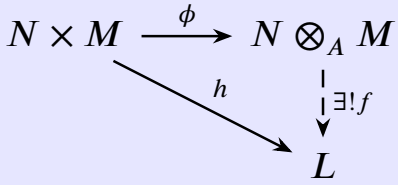
Definition 1.9. M *is an* ***A-module*** *if* $(M, +)$ *is an abelian group with a map* $A \times M \to M$ *such that the following holds (roughly, meaning it is linear):*

(i) $a(m_1 + m_2) = am_1 + am_2$;
(ii) $(a_1 + a_2)m = a_1 m + a_2 m$;
(iii) $a_1(a_2 m) = (a_1 a_2)m$.
(iv) $1_A m = m$.

Further denote by $\underline{Mod}_A$ *the category of A-modules.*

1.3.2 Tensor Product

Definition 1.10. ***Tensor product*** $\bullet \otimes_A \bullet$ *is a bifunctor* $\underline{Mod}_A \times \underline{Mod}_A \to \underline{Mod}_A$ *satisfying the following universal property:*



where e *is a bilinear and* f *is an* A*-linear morphism.*

Definition 1.11. *Let $A \xrightarrow{f} B$ be a morphism of rings and let N be a B-module. Then, N can be viewed as an A-module where for $n \in N$, multiplication with $a \in A$ is defined as $f(a)n$. This is called a* ***restriction of scalars****.*

Definition 1.12. *Again, let $A \xrightarrow{f} B$ be a morphism of ring and now let M be an A-module. An* ***extension of scalars*** *is an B-module $M_B := B \otimes_A M$ where B is naturally perceived as an A-module and $b(b' \otimes x) = bb' \otimes x$. Note that extension of scalars is the coproduct of A-algebras.*

Definition 1.13. *Let $S \subseteq A$ be a multiplicatively closed and M an A-module. Let $K := \{m \in M : \exists_{s \in S} sm = 0\}$. Define the $A[S^{-1}]$-module $M[S^{-1}]$ by considering the pairs $\{m/s : m \in M/K, s \in S\}$, where $(m, s) \sim (m', s')$ iff $s'm = sm'$ and introducing the module structure as $\frac{a}{s_1} \cdot \frac{m}{s_2} := \frac{am}{s_1 s_2}$.*

Proposition 1.14. *Let $S \subseteq A$ be a multiplicative system and M an A-module. Then, $M[S^{-1}] \cong A[S^{-1}] \otimes_A M$.*

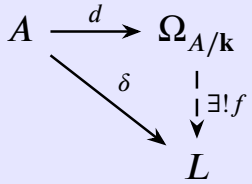
Definition 1.15. *M is a* ***flat*** *A-module if $(\cdot) \otimes_A M$ is left-exact.*

1.3.3 Differentials

In this section, we denote our commutative algebra by $\mathbf{k}$ (stressing that this is typically a field).

Definition 1.16. *Let A be a $\mathbf{k}$-algebra and M be an A-module. A $\mathbf{k}$-linear derivation $\delta : A \to M$ is a $\mathbf{k}$-linear map satisfying the Leibniz rule $\delta(a_1 a_2) = a_1\delta(a_2) + a_2\delta(a_1)$ for every $a_1, a_2 \in A$.*

Definition 1.17. *Let A be a $\mathbf{k}$-algebra. Then there exists a pair (Ω, d) of an A-module Ω and a $\mathbf{k}$-linear derivation $d : A \to \Omega$ that is a universal in the sense of the following diagram:*



Moreover, Ω is generated by the set $d(A)$.

Lemma 1.18. *Let $S = \mathbf{k}[T_1, \dots, T_n]$ and let $I = (f_1, \dots, f_r)$ be an ideal of S. Let $A = S/I$. Then,*

$$\Omega_{A/\mathbf{k}} \cong \frac{\Omega_{S/\mathbf{k}}}{I\Omega_{S/\mathbf{k}} + S(d(i) : i \in I)}.$$

Example. *Let us compute $\Omega_{\mathbb{C}/\mathbb{R}}$. In the setting of lemma above, $S = \mathbb{R}[T]$, $I = (T^2 + 1)$, and $A = \mathbb{R}[T]/(T^2 + 1) \cong \mathbb{C}$. Thus:*

$$\begin{aligned}
\Omega_{\mathbb{C}/\mathbb{R}} &\cong \frac{\Omega_{\mathbb{R}[T]/\mathbb{R}}}{(T^2+1)\Omega_{\mathbb{R}[T]/\mathbb{R}} + \mathbb{R}[T](2TdT)} \\
&\cong \frac{\mathbb{R}[T]dT}{\mathbb{R}[T](T^2+1)dT + 2T\mathbb{R}[T]dT} \\
&\cong \frac{\mathbb{R}[T]dT}{((T^2+1),(2T))\mathbb{R}[T]dT} \cong \frac{\mathbb{R}[T]dT}{\mathbb{R}[T]dT} \cong 0
\end{aligned}$$

1.4 Integrality, Valuation Rings

1.4.1 Integral Elements and Closure

Definition 1.19. *Consider a ring homomorphism $f : A \to B$. We say that $b \in B$ is **integral** over A if there exists a monic $f \in A[T]$ such that $f(b) = 0$. f is **integral** if every element of B is integral over A.*

Lemma 1.20. *If f is finite, then it is integral. Conversely, if B is finitely generated as an A-algebra by elements integral over A, then B is a finite A-module.*

Proof. Suppose f is finite, then $B = (b_1, \dots, b_n)$. Define $g_i : B \to B$ by multiplication by b_i. Then, by Cayley-Hamilton trick there exist $a_0, \dots, a_{n-1} \in A$ such that $b_i^n + a_{n-1}b_i^{n-1} + \cdots + a_0 = 0$, meaning that b_i is a root of some $A[T]$. Repeat for each b_i.

For the second part, suppose B is finitely generated by $b_1, \dots, b_n$ integral over A. Fix N large enough and write $b_i^N = \sum_{j=0}^{N-1} a_{ij}b_i^j$ for every i. By assumption, the A-module B is generated by the set $\{b_1^{s_1} \dots b_n^{s_n} : s_1, \dots, s_n \in \mathbb{Z}_{\geq 0}\}$. Using the polynomial relation above, we see that in fact the A-module is generated by $\{b_1^{s_1} \dots b_n^{s_n} : 0 \leq s_1, \dots, s_n < N\}$. □

Definition 1.21. *Let $A \to B$ be a ring homomorphism. Then, the set*

$$C := \{b \in B : b \text{ is integral over } A\}$$

is an A-subalgebra of B. It is called the integral closure of A in B.

Definition 1.22. *The **normalization** of a domain A is its integral closure in $\mathrm{Frac}(A)$. We say that A is normal if it is equal to its normalization.*

1.4.2 Valuation Rings

Definition 1.23. *Let K be a field. A **valuation subring** of K is a subring $K^+ \subseteq K$ such that for every non-zero $x \in K$, either $x \in K^+$ or $x^{-1} \in K^+$.*

Definition 1.24. *Let K be a field. A **valuation** on K is a map $|\cdot| : K \to \Gamma \cup \{0\}$ where Γ is a totally ordered abelian group such that $|0| = 0$, $|1| = 1$, $|xy| = |x||y|$, and $|x + y| \leq \max\{|x|, |y|\}$.*

1.5 Noetherian Rings

Definition 1.25. *Let A be a ring. The following conditions are equivalent:*

(i) Every ideal $I \subseteq A$ is finitely generated.

(ii) Every submodule $N \subseteq M$ of a finitely generated A-module M is finitely generated.

(iii) Every increasing sequence $I_0 \subseteq I_1 \subseteq \ldots$ of ideals in A stabilizes.

(iv) Every increasing sequence $N_0 \subseteq N_1 \subseteq \ldots$ of submodules of a finitely generated A-module M stabilizes.

*If either of the condition holds, A is a **Noetherian ring**.*

Proposition 1.26. *Let A be a Noetherian ring. Then the following hold.*

(a) For every ideal $I \subseteq A$, the ring A/I is Noetherian.

(b) Every localization of A is Noetherian.

*(c) **Hilbert's basis theorem.** The ring $A[T]$ is Noetherian.*

(d) Every finitely generated A-algebra is Noetherian and f.p. in A.

1.6 Graded Rings

Definition 1.27. *Let A be a ring and Λ be a commutative monoid. We say that A is Λ-**graded** if there exists a decomposition $A = \bigoplus_{\lambda\in\Lambda} A_\lambda$, where A_λ are abelian subgroups of A such that $1 \in A_0$ and $A_{\lambda_1} \cdot A_{\lambda_2} \subseteq A_{\lambda_1+\lambda_2}$.*
*We say M is a Λ-**graded** A-module if $M = \bigoplus_{\lambda\in\Lambda} M_\lambda$ such that $A_{\lambda_1} M_{\lambda_2} \subseteq M_{\lambda_1+\lambda_2}$ for every $\lambda_1, \lambda_2 \in \Lambda$.*

2 Results

2.1 Nilradical is the intersection of prime ideals

Lemma 2.1. $\mathfrak{N}_A = \bigcap_{\mathfrak{p}\in Spec(A)} \mathfrak{p}$.

Proof. Inclusion $\mathfrak{N}_A \subseteq \bigcap_{\mathfrak{p}\in \mathrm{Spec}(A)} \mathfrak{p}$ is easy. Suppose $x \in \mathfrak{N}_A$, meaning $x^n = 0$ for some n. So in particular, $x^n \in \mathfrak{p}$ for any prime ideal $\mathfrak{p} \in \mathrm{Spec}(A)$. If $n = 1$, we win. Otherwise, $x \cdot x^{n-1} \in \mathfrak{p}$, meaning either $x \in \mathfrak{p}$ or $x^{n-1} \in \mathfrak{p}$. If the former is true, we win. Otherwise, continue until we win. We eventually win.

Let us prove the other direction, which is a bit more tricky. Take any f that lies in every prime ideal. Consider the localization $A[f^{-1}]$. Since prime ideals of $A[f^{-1}]$ are in one-to-one correspondence to prime ideals $\mathfrak{p}$ of A with $\mathfrak{p} \cap \{f^n\}_{n\geq 0} = \emptyset$, $A[f^{-1}]$ contains no prime ideals, so $A[f^{-1}] = 0$. But then $1/1 = 0/1$ in $A[f^{-1}]$, meaning there is some $s \in S$ such that $s = 0$. But since $S = \{f^n\}_{n\geq 0}$, this proves the lemma. □

2.2 Spec(A) is quasi-compact

Lemma 2.2. *$X = Spec(A)$ is a quasi-compact topological space: meaning, every open covering of X has a finite subcovering.*

Proof. Let $X = \bigcup_{\alpha\in I} U_\alpha$. One might more concretely assume $X = \bigcup_{\alpha\in I} D(f_\alpha)$. Equivalently, if $I = (f_\alpha)_{\alpha\in I}$, then $V(I) = \emptyset$, so no prime ideal contains I. Thus, A/I is a zero-ring and so $A = I$. By definition of an ideal generated by a subset of a ring, we have a *finite* expression

$$1 = \sum_{\beta\in J} g_\beta f_\beta, \quad g_\beta \in A.$$

Thus $\bigcup_{\beta\in J} D(f_\beta)$ is the desired subcovering. □

2.3 Tensor Product is Right-exact

Proposition 2.3. *Let*

$$M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0 \quad (\star)$$

be an exact sequence in $\underline{Mod}_A$ *and let* $N \in \underline{Mod}_A$. *Then the sequence*

$$M' \otimes_A N \xrightarrow{f\otimes 1} M \otimes_A N \xrightarrow{g\otimes 1} M'' \otimes_A N \to 0$$

is exact.

We need the auxiliary fact to prove this proposition.

Proposition. *Let* $M' \to M \to M'' \to 0$ *be exact. This sequence is exact iff for all* $N \in \underline{Mod}_A$, *the sequence*

$$0 \to Hom(M'', N) \to Hom(M, N) \to Hom(M', N)$$

is exact. Similarly, $N' \to N \to N'' \to 0$ *is exact iff for all* $M \in \underline{Mod}_A$ *the following sequence is exact:*

$$0 \to Hom(M, N') \to Hom(M, N) \to Hom(M, N'').$$

We consider it to be known from basic category theory.

Proof of proposition. Let $P \in \underline{\mathrm{Mod}}_A$. Since sequence $(\star)$ is exact, the sequence $\mathrm{Hom}((\star), \mathrm{Hom}(N, P))$ is exact by proposition above. Since $\mathrm{Hom}(M \otimes_A N, P) \cong \mathrm{Hom}(M, \mathrm{Hom}(N, P))$, $\mathrm{Hom}((\star) \otimes_A N, P)$ is exact. By invoking proposition above once again, $(\star) \otimes_A N$ is exact. □

2.4 Structure of modules over PID

This theorem does not require knowing the proof for an exam.

Theorem 2.4. *Let* A *be a PID and let* M *be finitely generated* A*-module. Then,*

$$M \cong A^r \oplus \bigoplus_{i=1}^{s} A/(f_i^{m_i})$$

for some $r \geq 0$, *irreducible elements* $f_1, \ldots, f_s \in A$, *and* $m_1, \ldots, m_s \geq 1$.

2.5 Ring is local iff non-units form an ideal

Lemma 2.5. *A is local if any only if* $A \setminus A^\times$ *is an ideal.*

Proof. Suppose $(A, \mathfrak{m})$ is local. Let x be an element not in $\mathfrak{m}$. Then x must be a unit. Otherwise, x generates an ideal (x) which is contained in the maximal ideal other than $\mathfrak{m}$.

Conversely, suppose $I = A \setminus A^\times$ is an ideal. Then every ideal of A must be contained in I since ideals cannot contain units. □

2.6 Nakayama's Lemma

Lemma 2.6. *Let $(A, \mathfrak{m})$ be a local ring and let M be a finitely generated A-module. Then, $M = \mathfrak{m}M$ implies $M = 0$.*

Proof. Let $M = (x_1, \dots, x_n)$. Since $\mathfrak{m}M = M$, then

$$x_i = \sum_{j=1}^{n} m_{ij} x_j \text{ for some } m_{ij} \in \mathfrak{m}.$$

Thus $\mathbf{x} = Q\mathbf{x}$ for some $Q = \{m_{ij}\}$ and so $(E_{n\times n} - Q)\mathbf{x} = 0$. Consider the matrix $T = E_{n\times n} - Q$. Let $\pi : A \to A/\mathfrak{m}$ be the natural projection, then $\pi(T) = E_{n\times n}$, and so $\det \pi(T) = 1$ in $\mathfrak{m}$. Since $\det \pi(T)$ is invertible in $A/\mathfrak{m}$, it is not in $\mathfrak{m}$, and thus invertible in A. Thus, T is invertible as well. Thus, $\mathbf{x} = (m_1, \dots, m_n) = 0$. □

2.7 Finitely presented flat module over local ring is free

Lemma 2.7. *Let A be a local ring and M is a finitely presented flat module. Then, M is free.*

Proof. If M is finitely presented, then

$$0 \to K \to A^n \to M \to 0$$

is exact and K is finitely generated. Apply $\bullet \otimes_A (A/\mathfrak{m})$ functor. Suppose elements m_i generate M, then their images generate $M/\mathfrak{m}M$ as $R/\mathfrak{m}$-vector space. We may assume these images $\overline{m}_i$ actually form a basis. Therefore, the map $(A/\mathfrak{m}A)^n \to M/\mathfrak{m}M$ is an isomorphism. However, in such case $K/\mathfrak{m}K \cong 0$ and so by Nakayama $K = 0$. Thus, $M \cong A^n$.

2.8 Going-up Theorem

Theorem 2.8. *Suppose $f : A \to B$ is an integral ring homomorphism, then the induced map $f^* : Spec(B) \to Spec(A)$ is closed.*

Proof. Let $Z = V(J) \subseteq \mathrm{Spec}(B)$ be a closed subset for some ideal $J \subseteq B$. We must show that $f^*(V(J))$ is closed. Since this is an image of the composition $\mathrm{Spec}(B/J) \to \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ coming from the integral ring homomorphism $A \to B/J$, we might reduce our statement to $f^*(\mathrm{Spec}(B))$ must be closed. Clearly,

$\operatorname{im}(f^*) \subseteq V(\ker f)$. We shall prove equality. Recall that $V(\ker f)$ is the image of $\operatorname{Spec}(A/\ker f) \to \operatorname{Spec}(A)$. We factor $A \to B$ as $A \twoheadrightarrow (A/\ker f) \hookrightarrow B$. Replacing A by $A/\ker f$ we reduce to the case when $A \to B$ is injective and to the claim that $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective.

Let $\mathfrak{p} \in \operatorname{Spec}(A)$. The map $A_{\mathfrak{p}} \to B_{\mathfrak{p}}$ is integral and injective. In particular, $B_{\mathfrak{p}} \neq 0$. Take a maximal ideal $\mathfrak{m} \in \operatorname{Spec}(B_{\mathfrak{p}})$ and its image $\mathfrak{q} \in \operatorname{Spec}(A_{\mathfrak{p}})$. We have the following situation.

$$\begin{array}{ccc} A_{\mathfrak{p}} & \xrightarrow{\text{integral}} & B_{\mathfrak{p}} \\ \downarrow & & \downarrow \\ A_{\mathfrak{p}}/\mathfrak{q} & \hookrightarrow & B_{\mathfrak{p}}/\mathfrak{m} \end{array}$$

The composition $A_{\mathfrak{p}} \to B_{\mathfrak{p}}/\mathfrak{m}$ is integral, so also $A_{\mathfrak{p}}/\mathfrak{q} \hookrightarrow B_{\mathfrak{p}}/\mathfrak{m}$ is integral. But note that since $A_{\mathfrak{p}}/\mathfrak{q}$ is a field, so is $B_{\mathfrak{p}}/\mathfrak{m}$, so $\mathfrak{m}$ is maximal. Hence $\mathfrak{q}$ is a maximal ideal of $A_{\mathfrak{p}}$. This ring is local, so that $\mathfrak{q} = \mathfrak{p}A_{\mathfrak{p}}$. This implies that $\mathfrak{p} \in \operatorname{im}(f^*)$, which concludes the proof.

2.9 Artin-Tate Lemma

Lemma 2.9. *Let A be a Noetherian ring, let C be a finite type A-algebra, and let $B \subseteq C$ be an A-subalgebra such that C is finite over B. Then B is of finite type over A.*

Proof. Let $x_1, \ldots, x_m$ generate C as an A-algebra and let $y_1, \ldots, y_n \in C$ generate C as B-module. Then,

$$x_i = \sum_j b_{ij} y_j, \quad y_i y_j = \sum_k b_{ijk} y_k$$

Let $A' \subseteq B$ be an A-algebra generated by these elements. Then $A \to A'$ is of finite type by definition. In particular, A' is Noetherian since A is. On the other hand, the above equations certify that $A' \to C$ is finite. Since B is an A'-submodule of C, and A' is Noetherian, we see that $A' \to B$ is finite, and in particular of finite type. Therefore $A \to A' \to B$ is of finite type. □

2.10 Hilbert's Basis Theorem

Theorem 2.10. *If A is Noetherian, so is $A[T]$.*

Proof. Let $J \subseteq A[T]$ be an ideal. Suppose first that J contains some monic polynomial f of degree n. Then, any $g \in J$ of degree $\deg(g) = m > n$ can be written as:

$$g = a_m T^{m-n} f + h$$

where h is of degree less than m. By induction, we show that J is generated by f and $J \cap \bigoplus_{i<n} A \cdot T^n$. The latter is an A-submodule of $\bigoplus_{i<n} A \cdot T^n$ which is finitely generated since A is Noetherian. If follows that J is finitely generated.

Assume we do not have any monic polynomials in J. Let $I \subseteq A$ be the set consisting of 0 and all possible leading coefficients of elements in J. It is an ideal in A, and hence is finitely generated, say $I = (a_1, \dots, a_n)$ with $a_i \neq 0$. By definition, we have elements $f_i = a_i T^{n_i} + \cdots \in J$ with leading coefficient a_i. Let $N := \max_i\{n_i\}$, then we similarly show that J is generated by $(f_1, \dots, f_n)$ together with $J \cap \bigoplus_{i<n} A \cdot T^n$, and hence is finitely generated as A is Noetherian. □

2.11 Nullstellensatz Lemma

Lemma 2.11. *Let* $\mathbf{k}$ *be a field and* L *a finitely generated* $\mathbf{k}$*-algebra. Then,*

(i) If $L/\mathbf{k}$ *is a field extension,* $[L : \mathbf{k}] < \infty$.

(ii) Let $\mathfrak{m} \subseteq L$ *be a maximal ideal, then* $[L/\mathfrak{m} : \mathbf{k}] < \infty$. *In particular, if the field is algebraically closed (*$\mathbf{k} = \overline{\mathbf{k}}$*), then* $L/\mathfrak{m} \cong \mathbf{k}$.

(iii) If $\mathbf{k} = \overline{\mathbf{k}}$*, then* $MSpec(L) \cong Hom_{\mathbf{k}}(L, \mathbf{k})$.

(iv) Let $\mathbf{k} = \overline{\mathbf{k}}$ *and* $\mathbf{k}^n$ *is endowed with the Zariski topology: a set* $Z \subseteq \mathbf{k}^n$ *is closed iff it is of form*

$$V_k(I) = \{(x_1, \dots, x_n) \in \mathbf{k}^n : \forall_{f \in I} f(x_1, \dots, x_n) = 0\}$$

for some ideal $I \subseteq \mathbf{k}[T_1, \dots, T_n]$. *Then,* $Z \mapsto Z \cap \mathbf{k}^n$ *is a bijection between closed sets of* $Spec(L)$ *and* $Spec(\mathbf{k}^n)$.

(v) Define $I_k(Z) = \{f \in \mathbf{k}[T_1, \dots, T_n] : \forall_{\mathbf{x} \in Z} f(\mathbf{x}) = 0\}$. *Then,*

(a) For an ideal $I \subseteq \mathbf{k}[T_1, \dots, T_n]$, $I_k(V_k(I)) = \sqrt{I}$.

(b) For a subset $Z \subseteq \mathbf{k}^n$*, one has* $V_k(I_k(Z)) = \overline{Z}$.

(c) I_k *and* V_k *are mutually inverse bijections between the sets of radical ideals of* $\mathbf{k}[T_1, \dots, T_n]$ *and the set of Zariski closed sets of* $\mathbf{k}^n$.

2.12 Noether normalization lemma

Lemma 2.12. *Let* $\mathbf{k}$ *be a field and* A *a non-trivial finitely generated* $\mathbf{k}$*-algebra. Then, there exists an integral map* $k[y_1, \dots, y_n] \to A$ *where* $y_1, \dots, y_n$ *are algebraically independent.*

Proof. We prove this fact by induction. Base of induction is trivial. Assume $x_1, \dots, x_n$ are algebraically dependent and generate A, then there exists some polynomial f such that $f(x_1, \dots, x_n) = 0$ (otherwise $A \cong \mathbf{k}[x_1, \dots, x_n]$ and we are

done). Define new elements $z_1, \dots, z_{n-1}$ as follows: $z_i := x_i - x_n^{m_i}$ where m_i is some integer to be determined. Thus, we get:

$$f(z_1 + x_n^{m_1}, z_2 + x_n^{m_2}, \dots, z_{n-1} + x_n^{m_{n-1}}, x_n) = 0.$$

Now let $f = \sum_\alpha a_\alpha T_1^{\alpha_1} \dots T_n^{\alpha_n}$. When we expand the substitution, the typical term would be:

$$a_\alpha (z_1 + x_n^{m_1})^{\alpha_1} (z_2 + x_n^{m_2})^{\alpha_2} \dots (z_{n-1} + x_n^{m_{n-1}})^{\alpha_{n-1}} x_n^{\alpha_n}.$$

The highest power of x_n in such case is $x_n^{\alpha_n + \sum_{i<n} \alpha_i m_i}$. We choose the integers m_i such that all these exponents are distinct for every multi-index α where $a_\alpha \neq 0$ (for instance, let $m_i := r^i$ for large enough r). Since exponents are distinct, there is a unique maximum exponent N. The equation then takes the form $cx_n^N + (\dots) = 0$ where $c \in \mathbf{k}$. Dividing by c we get a monic equation of x_n with coefficients in $\mathbf{k}[z_1, \dots, z_{n-1}]$. Thus, x_n is integral over $\mathbf{k}[z_1, \dots, z_{n-1}]$, and thus so are x_i. Applying transitivity of integrality, we win.

2.13 Chevalley's Theorem

Theorem 2.13. *Let $f : A \to B$ be of finite presentation and let the induced continuous map be $f^* : Spec(B) \to Spec(A)$. Then, for every constructible subset $Z \subseteq Spec(B)$, its image $f^*(Z) \subseteq Spec(A)$ is constructible.*

2.14 Equivalent conditions for separable field extensions

Proposition 2.14. *The following statements are equivalent:*

(a) L/K *is separable.*
(b) $\Omega_{L/K} = 0$.
(c) $L \otimes_K \overline{K} \cong \overline{K}^n$ *for some* $n \geq 0$.
(d) $L \otimes_K L$ *is reduced.*
(e) $\#Hom_K(L, \overline{K}) = [L : K]$.
(f) $L \cong K[T]/(f(T))$ *for some monic irreducible* $f \in K[T]$.
(g) $Tr_{L/K} : L \to K$ *is non-zero.*
(h) $\langle x, y \rangle = Tr_{L/K}(xy) : L \times L \to K$ *is non-generate.*

(a) $\Rightarrow$ (b). Let $x \in L$ be an element with minimal polynomial $f \in K[T]$. Then, $\delta(f(x)) = f'\delta(x) = 0$ for every K-linear derivation $\delta : L \to L$. But since L/K is separable, $f' \neq 0$, and so $\delta = 0$.

(c) $\Rightarrow$ (d). Let $L \hookrightarrow \overline{K}$ be a natural inclusion. Tensor it with $\otimes_K$: $L \otimes_K L \hookrightarrow L \otimes_K \overline{K} \cong \overline{K}^n$. Since $\overline{K}^n$ is clearly reduced, so is $L \otimes_K L$.

2.15 Finiteness of integral closure

Theorem 2.15. *Let A be a domain with K its field of fractions. Let L be a finite field extension of K. Let $B \subseteq L$ be an integral closure of A in L:*

$$\begin{array}{ccc} B & \hookrightarrow & L \\ \uparrow & & \uparrow \\ A & \hookrightarrow & K \end{array}$$

Suppose that one of the following conditions hold:

(a) A is Noetherian and normal.

(b) A is of finite type over K.

(c) A is of finite type over $\mathbb{Z}$.

Then, $A \to B$ is finite. In particular, B is Noetherian.

2.16 Criterion for a graded ring to be Noetherian

Proposition 2.16. *Let $A = \bigoplus_{n \geq 0} A_n$ be a graded ring. A is Noetherian if and only if A_0 is Noetherian and A is of finite type over A_0.*

Proof. If A_0 is Noetherian and A is of finite type over A_0, then this statement follows from the Hilbert basis theorem.

Now, assume A is Noetherian. Let $A_+ = \bigoplus_{n>0} A_n$, then $A_0 = A/A_+$ and it is Noetherian since A is Noetherian. Now we show that $A_0 \to A$ is of finite type. Since A_+ is an ideal of a Noetherian ring, A_+ is finitely generated: say, $A_+ = (f_1, \dots, f_r)$. Without loss of generality, assume each f_i is homogeneous: say, $f_i \in A_{n_i}$. We claim that this set generates A as an A_0-algebra. We show this by induction. For $n = 0$ there is nothing to do. Let now $f \in A_n$ for $n > 0$. Then, $f \in A_+$, so we can write $f = \sum_i h_i f_i$ for $h_i \in A_{n-n_i}$. Since $n_i > 0$, we have $n - n_i < n$, by the induction assumption we have $h_i \in A_0[f_1, \dots, f_r]$, so $f \in A_0[f_1, \dots, f_r]$. $\square$